

SOFTWARE REQUIREMENT SPECIFICATION (SRS)

SOFTWARE DESIGN SPECIFICATION (SDS)

PHARMA ERP SYSTEM

DOCUMENT INFORMATION

Field	Details
Project Name	Pharma ERP System
Document Type	SRS + SDS
Version	1.0
Prepared For	Pharmaceutical Business Management System
Technology Stack	MERN Stack
Frontend	React.js
Backend	Node.js + Express.js
Database	Postgre
Authentication	JWT
Mobile Support	React Native
Deployment	Cloud/VPS

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1. INTRODUCTION

The Pharma ERP System is a complete enterprise-level software solution developed for pharmaceutical companies to automate and manage their entire business workflow.

The system integrates:

- Product Management
- Inventory Management
- Warehouse Management
- CRM
- Distributor Management
- Retailer Ordering
- Billing
- Accounting
- GPS Tracking
- Reporting
- Notifications
- Analytics

The system centralizes all operations into a single platform.

2. PURPOSE OF THE SYSTEM

The purpose of the Pharma ERP System is to:

- Digitize pharmaceutical business operations

- Reduce manual work
 - Improve inventory control
 - Automate billing and taxation
 - Monitor field employees
 - Improve customer relationship management
 - Provide real-time analytics
 - Improve dispatch and logistics
 - Reduce medicine expiry losses
 - Ensure GST compliance
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3. PROJECT SCOPE

The project scope includes:

Included Modules

1. Master Admin Panel
 2. Product Management
 3. Inventory & Warehouse Management
 4. C&F Management
 5. Distributor Portal
 6. Retailer Ordering System
 7. Wholesale Billing System
 8. Medical Representative Module
 9. GPS & Location Tracking
 10. CRM Module
 11. Accounting & Finance
 12. Alerts & Notifications
 13. Smart Analytics
 14. Future Export Readiness
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4. OBJECTIVES

Business Objectives

- Increase operational efficiency
- Improve sales monitoring
- Reduce inventory wastage
- Improve customer service
- Increase profit margin
- Enable business scalability

Technical Objectives

- Real-time system synchronization
 - Secure authentication
 - Cloud-ready architecture
 - Mobile accessibility
 - Scalable microservice-friendly backend
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5. STAKEHOLDERS

Stakeholder	Role
Admin	Complete system management
Warehouse Manager	Inventory operations
Distributor	Product ordering
Retailer	Retail purchases
Medical Representative	Field sales operations
Accountant	Financial operations
Management	Business analytics
Transport Team	Delivery management

6. FUNCTIONAL REQUIREMENTS

6.1 USER AUTHENTICATION

Features

- User Registration
- Secure Login
- JWT Authentication
- Role-Based Access Control
- Password Encryption
- Session Management
- Multi-device Login Support

Inputs

- Username
- Password
- Role

Outputs

- JWT Token
 - Session Information
-

6.2 MASTER ADMIN PANEL

Features

- Dashboard
- Sales Monitoring
- User Management
- State-wise Reports
- Notification Center
- Activity Logs

Functional Requirements

- Admin shall monitor all sales.
 - Admin shall track pending payments.
 - Admin shall generate analytics reports.
 - Admin shall manage users.
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6.3 PRODUCT MANAGEMENT MODULE

Features

- Product Creation
- Batch Management
- Expiry Tracking
- Barcode Management
- Pricing Management
- GST Management

Functional Requirements

- System shall store product details.
- System shall track batch information.
- System shall manage product pricing.
- System shall monitor medicine expiry.

Product Fields

Field	Description
Product Name	Medicine Name
Product Code	Unique Product ID
Batch Number	Manufacturing Batch
MRP	Maximum Retail Price
GST Percentage	Applicable Tax
Expiry Date	Expiry Information
Composition	Chemical Composition

6.4 INVENTORY & WAREHOUSE MANAGEMENT

Features

- Multi-location Inventory
- Stock Tracking
- Warehouse Transfer
- Low Stock Alerts
- Dead Stock Monitoring
- Expiry Monitoring

Functional Requirements

- System shall update inventory automatically.
- System shall track inward/outward stock.
- System shall generate low stock alerts.
- System shall track warehouse transfers.

Workflow

Product Creation → Batch Allocation → Warehouse Storage → Dispatch → Sales

6.5 C&F MANAGEMENT

Features

- C&F Login
- Dispatch Tracking
- LR Tracking
- Challan Generation
- Delivery Monitoring

Functional Requirements

- System shall allow C&F access.
 - System shall track dispatch operations.
 - System shall generate delivery reports.
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6.6 DISTRIBUTOR PORTAL

Features

- Distributor Login
- Product Catalog
- Online Ordering
- Outstanding Tracking
- Invoice Download
- Dispatch Tracking

Functional Requirements

- Distributor shall place orders.
 - Distributor shall view invoices.
 - Distributor shall track order status.
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6.7 RETAILER ORDERING SYSTEM

Features

- Retailer Login
- Product Browsing
- Reorder Functionality

- Payment Tracking
- Offer Visibility

Functional Requirements

- Retailers shall browse products.
 - Retailers shall place orders.
 - Retailers shall track payments.
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6.8 WHOLESALE BILLING SYSTEM

Features

- GST Billing
- E-Way Bill Support
- E-Invoice Support
- Barcode Billing
- Credit Notes
- Sales Returns
- Expiry Returns
- Multi-rate Billing

Functional Requirements

- System shall generate GST invoices.
- System shall support barcode scanning.
- System shall handle returns.
- System shall generate e-invoices.

Billing Workflow

Customer Selection → Product Selection → Batch Selection → Invoice Generation → Stock Update → Payment Entry

6.9 MR MODULE

Features

- Doctor Visits
- Chemist Visits
- Order Booking
- Daily Reporting

- Tour Planning
- Meeting Scheduling

Functional Requirements

- MR shall record doctor visits.
 - MR shall submit daily reports.
 - MR shall track targets.
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6.10 GPS & LOCATION TRACKING

Features

- GPS Attendance
- Geo-tagging
- Check-in/Check-out
- Daily Movement Tracking
- Route History
- Territory Tracking

Functional Requirements

- System shall capture GPS coordinates.
 - System shall verify field visits.
 - System shall track employee movement.
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6.11 CRM MODULE

Features

- Lead Management
- Follow-up Management
- Meeting Scheduling
- Sales Pipeline Tracking
- Distributor Onboarding

Functional Requirements

- System shall create leads.
 - System shall assign leads.
 - System shall track conversion.
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6.12 ACCOUNTING & FINANCE

Features

- Ledger Management
- Outstanding Tracking
- Profit & Loss
- GST Reports
- Bank Reconciliation

Functional Requirements

- System shall generate financial reports.
 - System shall track payments.
 - System shall generate GST reports.
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6.13 ALERTS & NOTIFICATIONS

Features

- Payment Reminders
- Expiry Alerts
- Dispatch Notifications
- Meeting Reminders
- Low Stock Alerts

Functional Requirements

- System shall send automated alerts.
 - System shall notify users in real-time.
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7. NON-FUNCTIONAL REQUIREMENTS

Performance

- Response time shall be below 3 seconds.
- System shall support 500+ concurrent users.

Security

- JWT Authentication
- Role-based authorization
- Password hashing using bcrypt
- HTTPS communication

Scalability

- Modular architecture
- Cloud deployment support
- Horizontal scaling support

Reliability

- Daily database backup
- Error logging system
- Failover recovery

Availability

- 99.9% uptime

Compatibility

- Mobile support
- Desktop support
- Tablet support

8. USER ROLES & PERMISSIONS

Role	Permissions
Super Admin	Full system access
Admin	Operational management
Warehouse Manager	Inventory management
Accountant	Financial management
Distributor	Product ordering

Role	Permissions
Retailer	Retail ordering
MR	Visit tracking
Transport Staff	Delivery management

9. SYSTEM ARCHITECTURE

Architecture Type

Three-Tier MERN Architecture

```
Frontend (React.js)
  ↓
Backend API (Node.js + Express.js)
  ↓
Database (Postgre)
```

Components

Frontend Layer

- React.js
- Redux Toolkit
- Tailwind CSS
- Axios

Backend Layer

- Express.js
- JWT Authentication
- REST APIs
- Middleware

Database Layer

- Postgre
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10. DATABASE DESIGN

Main Collections/Tables

Collection	Purpose
users	User details
products	Product information
batches	Batch details
inventory	Stock tracking
invoices	Billing records
invoiceItems	Invoice products
distributors	Distributor details
retailers	Retailer details
gpsLogs	GPS tracking
visits	MR visits
leads	CRM leads
notifications	Alert system
payments	Payment records
warehouses	Warehouse management

11. API DESIGN

Authentication APIs

| Method | Endpoint | Purpose | |---|---| | POST | /api/auth/register | User registration | | POST | /api/auth/login | User login | | GET | /api/auth/profile | User profile |

Product APIs

Method	Endpoint
POST	/api/products

Method	Endpoint
GET	/api/products
PUT	/api/products/?:id
DELETE	/api/products/?:id

Inventory APIs

Method	Endpoint
GET	/api/inventory
POST	/api/inventory/transfer

Billing APIs

Method	Endpoint
POST	/api/invoices
GET	/api/invoices
POST	/api/returns

GPS APIs

Method	Endpoint
POST	/api/gps/checkin
POST	/api/gps/checkout
GET	/api/gps/history

12. UI DESIGN

Dashboard Design

Admin Dashboard

Widgets:

- Total Sales
- Total Orders
- Pending Payments
- Live Inventory
- Top Products
- Revenue Charts

Distributor Dashboard

Widgets:

- Orders
- Outstanding Amount
- Invoice Downloads
- Dispatch Status

MR Dashboard

Widgets:

- Today's Visits
 - Targets
 - Attendance
 - Route Map
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13. MODULE DESIGN

Product Module Design

Components

- Product Form
- Batch Form

- Barcode Generator
- Pricing Module

Inventory Module Design

Components

- Warehouse View
- Stock Table
- Transfer Module
- Alert System

Billing Module Design

Components

- Invoice Generator
 - GST Calculator
 - Barcode Scanner
 - Return Processing
-

14. SECURITY DESIGN

Authentication

- JWT-based Authentication
- Access Tokens
- Refresh Tokens

Authorization

- Role-Based Access Control

Encryption

- bcrypt password hashing

Data Protection

- HTTPS

- Secure APIs
 - Input validation
 - XSS protection
 - SQL/NoSQL injection prevention
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15. GPS TRACKING DESIGN

Components

- Mobile GPS SDK
- Geo-tagging Module
- Attendance Tracker
- Route Mapping

Workflow

Login → GPS Enable → Check-In → Visit Tracking → Check-Out → Route History

Technologies

- Google Maps API
 - React Native
 - Socket.io
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16. BILLING SYSTEM DESIGN

Components

- Invoice Engine
- GST Calculator
- Barcode Scanner
- E-Way Bill Generator
- Return Management

Invoice Flow

Order → Invoice → GST → Payment → Dispatch → Delivery

Reports

- Sales Reports
 - GST Reports
 - Outstanding Reports
-

17. CRM DESIGN

CRM Workflow

Lead → Follow-Up → Meeting → Proposal → Conversion

Components

- Lead Form
- Follow-Up Calendar
- Meeting Scheduler
- Sales Pipeline

Features

- Lead Assignment
 - Lead Tracking
 - CRM Analytics
-

18. INVENTORY DESIGN

Inventory Features

- Batch Tracking
- Expiry Monitoring
- Low Stock Alerts
- Warehouse Transfers

Inventory Workflow

Product Inward → Batch Allocation → Storage → Dispatch → Sales

19. REPORTING SYSTEM

Reports

Report	Description
Sales Report	Sales analytics
Inventory Report	Stock analytics
GST Report	Tax filing
MR Report	Field performance
Outstanding Report	Pending payments
Dispatch Report	Shipment analytics

Export Formats

- PDF
 - Excel
 - CSV
 - png/jpg
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20. NOTIFICATION SYSTEM

Notification Types

- SMS Alerts
- Email Alerts
- Push Notifications
- In-App Notifications

Trigger Events

- Low Stock
 - Payment Due
 - Meeting Reminder
 - Dispatch Update
 - Expiry Alert
 - Out of stock
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21. FUTURE ENHANCEMENTS

Planned Features

- AI-based analytics
 - Demand forecasting
 - Mobile app expansion
 - Manufacturing integration
 - Export management
 - Multi-currency support
 - Voice assistant support
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22. TESTING STRATEGY

Testing Types

Testing	Purpose
Unit Testing	Individual components
Integration Testing	Module integration
System Testing	Complete system testing
Performance Testing	Load handling
Security Testing	Vulnerability analysis
User Acceptance Testing	Client approval

Testing Tools

- Postman
 - Jest
 - Mocha
 - Selenium
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23. DEPLOYMENT ARCHITECTURE

Deployment Environment

Frontend

- Vercel
- Netlify

Backend

- VPS Server
- AWS EC2
- DigitalOcean

Database

- Postgre

CI/CD

- GitHub Actions
 - Docker
 - Nginx
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24. HARDWARE REQUIREMENTS

Server Requirements

Component	Minimum
CPU	4 Core
RAM	8 GB
Storage	100 GB SSD
Internet	High-Speed

Client Requirements

Component	Minimum
Browser	Chrome/Edge
RAM	4 GB
Internet	Broadband

25. SOFTWARE REQUIREMENTS

Software	Purpose
Node.js	Backend Runtime
MongoDB	Database
React.js	Frontend
Express.js	API Server
VS Code	Development IDE
GitHub	Version Control
Postman	API Testing

26. CONCLUSION

The Pharma ERP System is a complete enterprise-level business automation solution for pharmaceutical organizations.

The system integrates:

- Inventory Management
- Billing
- CRM
- GPS Tracking
- Finance
- Warehouse Management
- Distribution Management
- Analytics

into a centralized digital platform.

The solution improves:

- Operational efficiency
- Sales management
- Employee monitoring
- Customer relationship management
- Business scalability
- Compliance management

The MERN-stack-based architecture provides:

- Scalability
- Security
- High performance
- Cloud readiness
- Mobile accessibility

making the system suitable for large-scale pharmaceutical business operations.